

| Question |     |     | Marking Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks Available |          |          |           |          |          |
|----------|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | AO1             | AO2      | AO3      | TOTAL     | Maths    | Prac     |
| 1        | (a) | i   | 8 (ms) = 2 marks<br>Award 1 mark for:<br>0.0083<br>1/120<br>8.3<br>8.33                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 | 2        |          | 2         | 2        |          |
|          |     | ii  | <u>Threshold</u> was not reached (so no action potential) (1)<br>Not enough {sodium ions/ Na <sup>+</sup> } entered the {neurone/ axon}(1)<br>Reject sodium unqualified                                                                                                                                                                                                                                                                                                                                          |                 | 2        |          | 2         |          |          |
|          | (b) | i   | Schwann (cell)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1               |          |          | 1         |          |          |
|          |     | ii  | Lipid/ fat                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1               |          |          | 1         |          |          |
|          |     | iii | $5.15 \times 10^{-5}$ s = 3 marks<br>0.0000515 s = 2 marks<br>$5\,150/1 \times 10^8$ = 1 mark                                                                                                                                                                                                                                                                                                                                                                                                                    |                 | 3        |          | 3         | 3        |          |
|          | (c) | i   | Any <b>four</b> (×1) from<br>A. Rate would increase (1)<br>B. Because more <u>aerobic</u> respiration (1)<br>C. Saltatory conduction would not occur/<br>Whole membrane would need to be depolarised/<br>action potentials would occur at more places along neurone/<br>shorter local circuits occur (1)<br>D. so more {Na <sup>+</sup> /K <sup>+</sup> } pumps (have to work across whole length of<br>exposed membrane)/<br>more active transport(1)<br>E. More ATP needed (to maintain resting potential) (1) |                 |          | 4        | 4         |          |          |
|          |     | ii  | Any <b>two</b> (×1) from:<br>Person would feel tired/ exhaustion/ feel weaker (1)<br>Lack of feeling/ OWTTE (1)<br>{Longer/ slower} reaction times (1)<br>paralysis/ no response/ loss of muscle control (1)                                                                                                                                                                                                                                                                                                     |                 |          | 2        | 2         |          |          |
|          |     |     | <b>QUESTION 1 TOTAL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>2</b>        | <b>7</b> | <b>6</b> | <b>15</b> | <b>5</b> | <b>0</b> |